

Q1. Explain the different types of data used in cluster analysis and discuss each type's effect on choice of clustering method.

Ans. Numerical Data

- Interval Data - Temp., marks, income
- Ratio Data - age, weight, height

Categorical Data:

- Nominal Data - No order
- Ordinal Data - Has order but no fixed difference

Binary Data: Only zero and one

Q2. ~~Diff~~ Describe the major categories of clustering methods. Compare partitioning, hierarchical, density based, grid based and model based approaches.

Ans. Partitioning Method (k-means)

- Divide data into k clusters

Hierarchical Methods:

- Create dendrogram (tree structure)

DBSCAN

- From cluster based on density

Grid-Based

- Divide spaces into grid

Model-Based Method:

- Assume ~~stat~~ statistical model

Q3. Discuss the working principle of hierarchical clustering. Differentiate b/w agglomerative and divisive methods with example.

1. Working Principle: Hierarchical clustering builds a tree structure called dendrogram.

merging or splitting clusters step-by-step.

Types.

1> Agglomerative (Bottom-up)

- Starts with each data point as a separate cluster.
- Merges closest clusters step-by-step.

2> Divisive (Top-Down)

- Starts with all data points in one cluster.
- Splits clusters step-by-step.

Q4> Define density-based clustering methods? Explain their advantages over partitioning methods and mention suitable applications.

Ans. Density-based clustering forms clusters based on dense regions of data points.

Advantages 1> Detects clusters of any shape.

2> Does not require predefined number of clusters.

3> Identifies noise and outliers.

Applications

- Spatial data analysis.
- Fraud detection.
- Image processing.

Q5> Explain constraint-based clustering and outlier analysis. How do constraints improve clustering results, and why is outlier detection important in data mining?

Ans. clustering method where user-defined constraints (such as distance, size, or rules) are applied to guide clustering.

- constraints improve accuracy and produce meaningful clusters.

Outlier Analysis:

Outliers are data points that are significantly different from others.

- Detects fraud or anomalies
- Improves data quality.
- Helps in better decision-making.